NICOLE MOOLMAN

20232190

APRIL 19, 2024

JD521 FORMATIVE ASSESMENT 2

Welcome to the Library Management System! This program allows librarians to manage books and members in a library. Below is a guide on how to use the system effectively.

How the Program Works:

1. **Book Class:**
    - Represents a book in the library system.
    - Includes attributes such as title, author, ISBN, and availability status.
    - Provides methods to toggle availability and retrieve book information.
2. **Member Class:**
    - Represents a member in the library system.
    - Includes attributes like name, email, and a list of borrowed books.
    - Validates email format and provides methods for borrowing and returning books.
3. **Library Class:**
    - Manages books and members in the library system.
    - Includes methods to add books and members, search for books, checkout and return books, and retrieve checked out books.
4. **LibraryManagementSystem Class:**
    - Main class for running the Library Management System.
    - Provides a menu-driven interface for librarians to perform various actions like adding books/members, searching books, checking out and returning books, and displaying information.

How to Run the Program:

1. **Compile:**
    - Compile the Java files using any Java compiler.
2. **Run:**
    - Run the compiled **LibraryManagementSystem** class.
    - Follow the on-screen prompts to navigate through the menu and perform desired actions.

Sample Usage:

1. Log in as a librarian with the password "admin".
2. Choose actions from the librarian menu, such as adding books/members, searching for books, checking out and returning books, and viewing checked out books.
3. Follow the prompts to input necessary information for each action.

Notes:

- For demonstration purposes, the password for the librarian is set to "admin". In real-world scenarios, ensure a more secure authentication method.
- This system operates in a command-line interface (CLI) environment

```
PS C:\Users\nicol>  & 'C:\Users\nicol\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.10.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\nicol\AppData\Local\Temp\vscodesws_06323\jdt_ws\jdt.ls-java-project\bin' 'LibraryManagementSystem'
Welcome to the Library Management System
1. Login as Librarian
2. Exit
Enter your choice: 1
Enter password: admin

Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice:
```

```
PS C:\Users\nicol>  & 'C:\Users\nicol\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.10.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\nicol\AppData\Local\Temp\vscodesws_06323\jdt_ws\jdt.ls-java-project\bin' 'LibraryManagementSystem'
Welcome to the Library Management System
1. Login as Librarian
2. Exit
Enter your choice: 1
Enter password: admin

Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice: 1
Enter title: Beginner Java
Enter author: Programmer
Enter ISBN: 12345678
Book added successfully.
```

```
Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice: 2
Enter member name: Mandy
Enter member email: Mandy@mail.com
Member added successfully.
```

```
Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice: 3

Books:
Title: Beginner Java, Author: Programmer, ISBN: 12345678
```

```
Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice: 4

Members:
Name: Mandy, Email: Mandy@mail.com
```

```
Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice: 5
Enter search query: Beginner Java

Search Results:
Title: Beginner Java, Author: Programmer, ISBN: 12345678
```

```
Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice: 6
Enter member email: Mandy@mail.com
Enter book ISBN: 12345678
Book checked out successfully.
```

```
Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice: 7

Checked Out Books:
Title: Beginner Java, Author: Programmer
```

```
Librarian Menu:
1. Add Book
2. Add Member
3. Display Books
4. Display Members
5. Search Books
6. Check Out Book
7. Display Checked Out Books
8. Return Book
9. Back to Main Menu
Enter your choice: 8
Enter member email: Mandy@mail.com
Enter book ISBN: 12345678
Book returned successfully.
```